Q1 logic

→ Read the value of N
→ Start a loop from 1 to N
→ Print each number followed by a space
→ stop when i becomes greater than N

Approach

→ use a for loop
→ Initialize i=1
→ continue while i<=N
→ Print i and increment it by 1.

Dry run

i	output
1	1
2	1 2
3	1 2 3
4	1 2 3 4
5	1 2 3 4 5

Time complexity

O(N) Because the loop run exactly N times.

Space complexity

O(1) only one loop variable (i) is used, so the extra memory is constant.

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Q2 logic :-

→ Read th value of N
→ initialize sum=0
→ use a loop from 1 to n
→ Add each number to sum
→ Print th final sum

Approach
→ Take input N
→ Initialize a variable sum to 0
→ Traverse from 1 to N using a for loop
→ Keep adding each number to sum
→ Display th result.

Dry run

i	sum
initial	0
1	1
2	3
3	6
4	10
5	15

Time complexity :- O(N), Th loop run exactly N times

Space complexity :- O(1), Two extra variables are used so, th ~~ext~~ extra memory is constant.

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03 logic

→ Read the value of A and b
→ initialize result = 1
→ run a loop from 1 to b
→ Multiply result by a in each iteration
→ Print result

Approach

→ Take input A and B
→ Start with result = 1
→ Repeat multiplications B time using a for loop
→ output the final value of result.

Dry run A=2 B=5

i	Result
initial	1
1	2
2	4
3	8
4	16
5	32

Time complexity :- O(B) The loop runs exactly B times

space complexity :- O(1) Only a constant amount of extra ~~memy~~ memory is used.